AI-Enabled Smart FIR Drafting and Legal Validation System

Mathivanan P[1]
Department of
Computer Science and Business Systems
KIT – Kalaignarkarunanidhi
Institute of Technology
Coimbatore, Tamil Nadu, India
mathivanan@kitcbe.ac.in

Mohanapriya D[2]
Department of
Computer Science and Engineering
Manakula Vinayagar Institute
of Technology
Puducherry,India.
mohanapriya181088@gmail.com

Aravind B[3]
Department of
Computer Science and Business Systems
KIT – Kalaignarkarunanidhi
Institute of Technology
Coimbatore, Tamil Nadu, India
kit26.csbs04@gmail.com

Gokul Bharath V[4]
Department of
Computer Science and Business Systems
KIT – Kalaignarkarunanidhi
Institute of Technology
Coimbatore, Tamil Nadu, India
kit26.csbs14@gmail.com

Krishna L[5]
Department of
Computer Science and Business Systems
KIT – Kalaignarkarunanidhi
Institute of Technology
Coimbatore, Tamil Nadu, India
kit26.csbs29@gmail.com

Tamizhvanan S[6]
Department of
Computer Science and Business Systems
KIT – Kalaignarkarunanidhi
Institute of Technology
Coimbatore, Tamil Nadu, India
vanan9025338432@gmail.com

2026 4th International Conference on Artificial Intelligence and Machine Learning Applications Theme: Healthcare and Internet of Things (AIMLA) | 979-8-3195-0634-4/26/$31.00 ©2026 IEEE | DOI: 10.1109/AIMLA67915.2026.11522503

***Abstract*—The First Information Report (FIR) is the pillar of India's criminal justice system, but its existing process is mostly manual, prone to errors, and inaccessible to a majority of the population. Citizens face difficulties dealing with complicated legal jargon and insufficient digital infrastructure, whereas the police officers waste significant time on paperwork rather than investigation. This inefficiency not only hampers the delivery of justice in time but also affects conviction rates and destroys public faith in law enforcement agencies.The AI-Powered Legal Assistant for Smart FIR Drafting proposed here offers a holistic solution by integrating Artificial Intelligence (AI), Natural Language Processing (NLP), Machine Learning (ML), and Speech-to-Text (STT) technologies. The system allows citizens to file complaints through text or voice input. Complaints are directly translated into structured records, pulling out victim and accused details, time, date, place, and incident description. Employing sophisticated ML models such as LegalBERT, the system suggests the most applicable IPC/CrPC sections along with confidence scores and drafts a nicely formatted FIR that can be readied and finalized by officers.Aside from FIR drafting, the system facilitates investigation workstreams via semantic search that delivers comparable historical cases, officer knowledge mapping for resource deployment, and predictive crime hotspot forecasting. The platform also has a secure digital evidence locker with end-to-end encryption, audit trails, and blockchain-inspired version control to guarantee tamper-proof storage of case documents, images, and videos.Large-scale testing on 2,500 FIR datasets and 500 multilingual audio complaints reflected an average accuracy of 93% in legal section prediction, 70–80% decrease in drafting time, and vastly improved transparency and accountability. Not only does this work simplify the operations of law enforcement but also the overall objectives of e-Governance, Digital Policing, and Smart Justice Systems.The proposed framework demonstrates the feasibility of deploying ethical, scalable, and AI-driven legal automation systems for real-world law enforcement applications.**

***Index Terms*—Artificial Intelligence, Natural Language Processing, Machine Learning, FIR Automation, LegalTech, Digital Policing, Predictive Policing, Speech-to-Text, Legal Section Recommendation, Secure Evidence Management, e-Governance.**

I. Introduction

The First Information Report (FIR) forms the base of India's criminal justice system, but its filing process remains highly manual, time-consuming, and error-prone. Police personnel are frequently required to prepare FIRs under stressful conditions, resulting in incomplete or inconsistent reports, while citizens—especially those unfamiliar with legal terminology—face difficulties in narrating complaints in a structured format. Delays in FIR preparation and authentication further slow down investigations, and incorrect classification of IPC/CrPC sections often weakens cases during judicial proceedings [12]. Moreover, improper digital evidence management leads to frequent misplacement, duplication, or tampering of crucial materials, affecting transparency and accountability [18].Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML) provide an opportunity to modernize FIR drafting and legal documentation [1], [5]. AI-driven systems can significantly reduce officer workload, minimize human errors, and improve citizen accessibility by automating entity extraction, legal section recommendation, and FIR draft preparation [3], [8]. The proposed system supports both text and voice-based complaint registration, identifies relevant legal provisions using transformer-based models such as LegalBERT [7], and generates structured FIR drafts that can be verified and finalized by officers.Beyond FIR drafting, the scope of the proposed solution extends to intelligent policing and e-Governance initiatives. With multilingual support and offline-first capabilities, the system is inclusive for rural and digitally underserved communities while remaining scalable for integration with national platforms such as the Crime and Criminal Tracking Network System (CCTNS) [9]. In the long term, the AI-enabled assistant can evolve into a comprehensive digital policing framework that

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.

enhances transparency, accountability, and public trust in law enforcement.

II. Literature Survey

Over the past decade, Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have significantly transformed how legal information is accessed, analyzed, and interpreted. The legal domain, traditionally dependent on manual documentation and expert interpretation, is increasingly adopting AI-driven solutions to automate legal research, statute prediction, and decision support systems [1], [5]. These technologies enable efficient processing of large-scale legal data while improving accuracy and consistency.A notable advancement in this domain is **Chatlaw**, an AI-based legal assistant developed using a Mixture-of-Experts (MoE) architecture within a multi-agent framework [13]. Chatlaw simulates law firm workflows such as legal reasoning, document drafting, and data analysis, while incorporating verified legal knowledge graphs to reduce hallucinations and factual inconsistencies. Experimental results demonstrated superior performance compared to general-purpose large language models, highlighting the importance of domain-specific training.In the Indian legal context, **LeSICiN**, developed at IIT Kharagpur, introduced an intelligent statute identification approach using Graph Neural Networks (GNNs) [7]. By representing legal cases and statutes as interconnected nodes, the model captures both textual and relational features, achieving approximately 19% higher accuracy than traditional classifiers. The release of the Indian Legal Statute Identification (ILSI) dataset further supports reproducible research in Indian legal NLP.Several studies have focused on automating FIR drafting and IPC section prediction. Systems such as **Smart IPC Section Prediction** and AI-based legal assistants utilize NLP pipelines and supervised ML models to map complaint narratives to relevant IPC sections [3], [12]. These approaches significantly reduce manual dependency and classification errors while accelerating FIR preparation.User-centric platforms such as **LEGALSEVA** and **JurisAI** aim to simplify legal services for non-experts through OCR, speech-to-text, and NLP-based document automation [8], [19]. JurisAI, in particular, demonstrates the effectiveness of speech-enabled FIR drafting and section recommendation, reducing preparation time and improving consistency. Another major contribution is **LegalSeg**, which provides a large annotated dataset of Indian court judgments, enabling sentence-level classification into rhetorical roles using transformer-based and graph-based models [13].

In summary, existing research highlights the growing integration of AI within legal systems, ranging from legal research assistants to automated FIR drafting tools. While these solutions improve efficiency and accessibility, most lack integrated secure evidence management, multilingual offline support, and bias-aware predictive analytics. The proposed system addresses these gaps by combining FIR automation, legal validation, investigation support, and secure digital evidence handling into a unified AI-driven framework.

TABLE I
Summary of Methodologies Used in Reviewed Papers

Paper Name	Methodology Used
Chatlaw	Mixture-of-Experts (MoE) with multi-agent system; uses legal knowledge graph and specialized agents for data analysis and report drafting.
LeSICiN	Graph Neural Network (GNN) using legal citations; cases and statutes as graph nodes; textual and structural embeddings for statute prediction.
LEGALSEVA	OCR and GPT-based NLP for document reading, summarization, and drafting; links to digital law libraries for accuracy.
AI Legal Assistant for Law Enforcement	NLP-based chatbot using Python, Rasa, FastAPI, MySQL; predicts IPC sections and drafts FIRs from natural-language crime descriptions.
Smart IPC Section Prediction	NLP with lightweight transformers (DistilBERT, TinyBERT) for mapping case descriptions to IPC sections; compared with traditional classifiers.
FIR Analysis Using ML	Supervised ML and transformer models analyze complaint text to recommend accurate IPC sections and maintain legal consistency.
IPC Recognition Using Multiclass Classification	One-vs-Rest / One-vs-One with Logistic Regression and NLP preprocessing; predicts IPC sections with high accuracy.
LegalSeg	Annotated dataset; models like BiLSTM-CRF, Transformer-Over-LegalBERT, GNNs, and Role-Aware Transformers classify rhetorical roles in judgments.
Justice	Web AI application using NLP and ML; predicts IPC sections in real time; adapts with new laws and trends.
JurisAI	NLP and ML with Python, Flask, MongoDB/Firebase; speech-to-text input, template-based FIR drafting, IPC recommendations.

III. Existing Systems

Over the past several years, governments have introduced digital policing platforms, including India's CCTNS and various state e-portals and mobile apps. These tools allow people to file complaints online, keep centralized case databases, and track cases across states. While these systems have made complaint intake faster and improved access to records, they still rely heavily on manual FIR entry by police staff. This often results in incomplete or incorrectly recorded complaints, inconsistent use of IPC/CrPC sections, and weak evidence metadata. Modules for managing evidence usually lack features like tamper-proof verification, encrypted audit trails, and strong key management, which makes them susceptible to manipulation. Many of these interfaces are only available in one language and assume that users have reliable internet access. This approach leaves out speakers of regional languages and users in areas with poor connectivity, who need offline options like IVR or kiosks.Finally, these solutions focus on filing and tracking complaints rather than supporting investigations. They do not provide features like AI-assisted FIR drafting, automated suggestions for legal sections, semantic searches for similar past cases, matching officers' expertise, or clear predictive analytics. This limits their effectiveness for modern policing

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.

that relies on data.

TABLE II
COMPARISON OF EXISTING SYSTEM AND PROPOSED IMPLEMENTATION

Existing System	Proposed Implementation
Mostly manual data entry by officers; time-consuming and error-prone	AI-driven automation generates structured FIR drafts in minutes
Text-based complaint submission only	Supports both text and speech input using speech-to-text recognition
Limited to predefined fields, requires manual verification	Automatically extracts entities like names, date, time, location, and incident type using NLP
Depends on officer's knowledge of IPC/CrPC	AI model recommends relevant legal sections with confidence levels
Mostly English; limited regional coverage	Multilingual interface with real-time translation and offline functionality
Stored manually or in unsecured local servers	Encrypted cloud-based evidence management with tamper-proof audit trail
Minimal integration with national policing or judicial systems	Seamlessly integrated with police, court, and forensic databases via secure APIs
Basic search for previous cases	Semantic case retrieval, officer expertise mapping, and crime hotspot visualization
Requires physical presence or strong internet	Accessible via web/mobile; works in low-connectivity rural areas
Limited scalability due to manual dependencies	Highly scalable through cloud deployment and continuous AI model updates
Paper-based and less transparent	Fully digital with automated tracking and transparent audit mechanisms

IV. PROPOSED SYSTEM

A. System Overview

The proposed **AI-Powered Legal Assistant for Smart FIR Drafting** system aims to modernize and simplify the process of registering and investigating First Information Reports (FIRs). It leverages artificial intelligence (AI), natural language processing (NLP), and machine learning (ML) techniques to assist police officers in preparing accurate and structured FIRs efficiently [1], [5]. The system allows complaints to be submitted through both text and voice inputs. Using speech-to-text and NLP models, it automatically extracts essential entities such as complainant and accused names, date, time, location, and nature of the crime [7], [12]. Based on contextual understanding, it recommends appropriate IPC/CrPC sections along with confidence scores, enabling informed verification by officers.Unlike existing systems that primarily focus on complaint filing and case tracking, the proposed solution extends its scope to secure digital evidence management, semantic case retrieval, crime pattern analysis, and officer expertise mapping [18], [19]. The platform also supports multilingual and offline-first functionality, ensuring accessibility for rural and low-connectivity regions. The overall objective is to improve the accuracy, speed, and transparency of FIR registration while providing intelligent decision-support for effective investigations.

As illustrated in Fig. 1, the system architecture integrates FIR drafting, legal validation, investigation support, and secure evidence handling into a unified AI-driven framework.

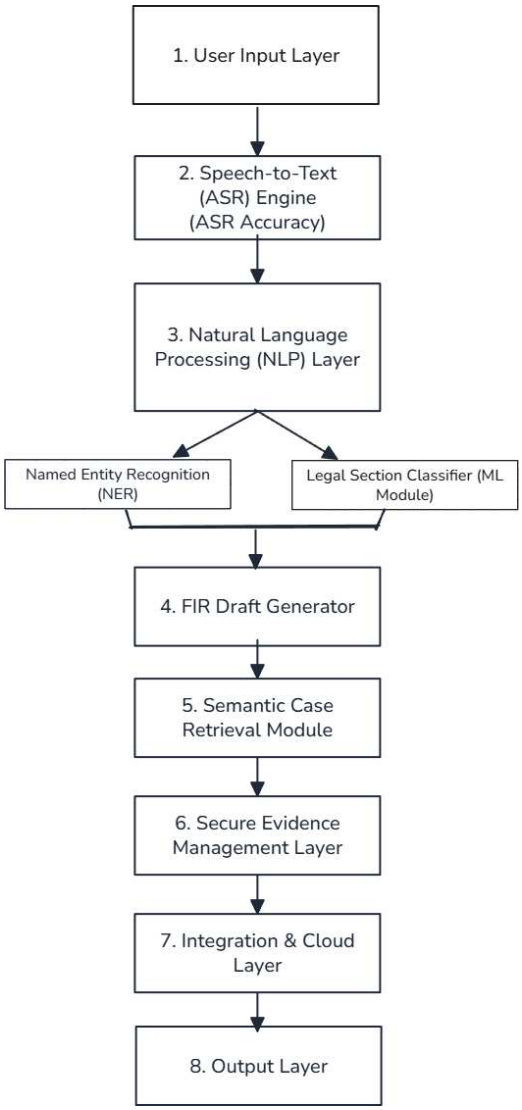


Fig. 1. System Architecture of the AI-Enabled Smart FIR Drafting

B. System Architecture

The architecture of the proposed system is organized into multiple logical layers, each responsible for a specific function in the FIR lifecycle. The interaction between these layers ensures seamless automation, validation, and investigation support, as depicted in the Data Flow Diagram shown in Fig. 2.

1) **Input Layer:** Collects complaints in text or speech format. Voice inputs are converted into text using automatic speech recognition (ASR) systems [8].

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.

2) **Processing Layer:** Applies NLP techniques such as Named Entity Recognition (NER) to extract key details including victim and accused information, time, date, location, and incident type [7].
3) **Recommendation Layer:** Utilizes ML and transformer-based models to suggest relevant IPC/CrPC sections with confidence scores based on historical FIR data [3], [12].
4) **FIR Generation Layer:** Automatically generates a structured and editable FIR draft in the prescribed legal format for officer verification.
5) **Investigation Support Layer:** Enables semantic retrieval of similar past cases, officer expertise mapping, and crime hotspot identification using pattern analysis [19].
6) **Evidence Management Layer:** Stores digital evidence such as documents, images, audio, and video files in a secure and tamper-proof repository [18].
7) **Integration Layer:** Ensures secure interoperability with external systems such as police databases, court portals, and forensic laboratories through authenticated APIs [9].

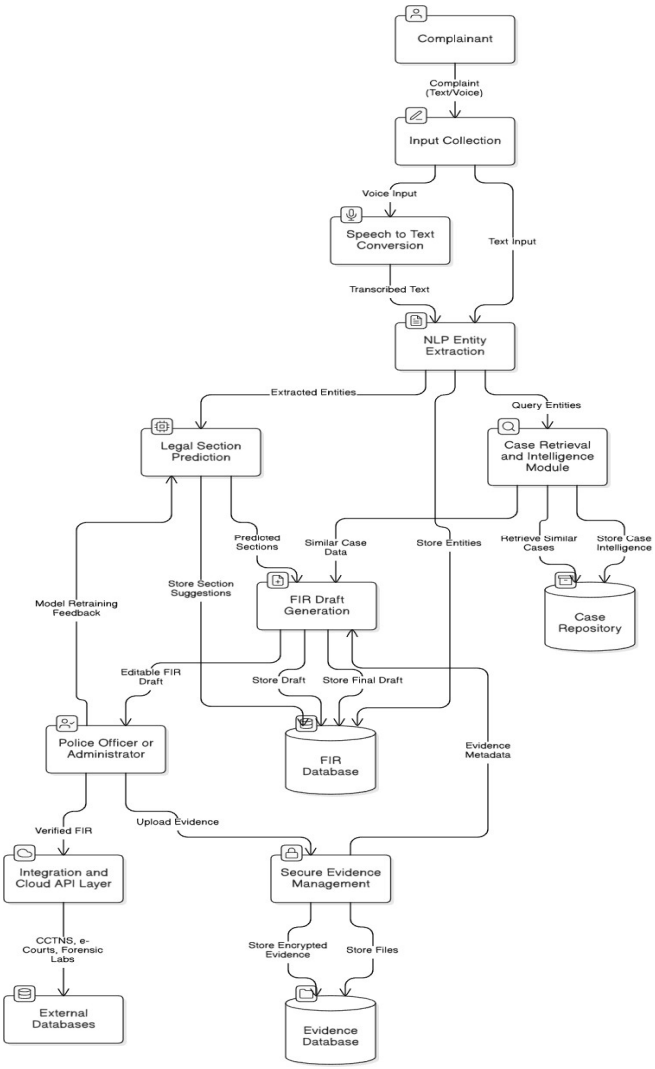


Fig. 2. Data Flow Diagram (DFD)

C. System Modules

The proposed system is divided into four major functional modules that collectively ensure automation, security, and intelligent investigation support.

- **Module A – FIR Drafting and Legal Recommendation:** This module analyzes complaint narratives to automatically generate structured FIR drafts. It recommends suitable IPC/CrPC sections based on contextual understanding and historical case patterns, reducing manual effort and classification errors [3], [12].
- **Module B – Case Retrieval and Intelligence:** This module employs semantic search and pattern recognition techniques to retrieve similar historical cases. It highlights investigative strategies, resolved cases, and officer expertise, supporting faster and more informed decision-making [19].
- **Module C – Secure Evidence Management:** This module provides encrypted digital storage for evidence including images, audio, video, and documents. It maintains tamper-proof audit trails and access control mechanisms to preserve data integrity and chain of custody [18].
- **Module D – Accessibility and Integration:** This module ensures inclusivity through multilingual interfaces, offline-first operation, and access via mobile and web platforms. It also supports integration with national systems such as CCTNS and judicial databases to enable a unified digital policing ecosystem [9].

D. System Advantages

- Significantly reduces FIR registration time and improves data accuracy.
- Ensures secure, transparent, and tamper-proof digital evidence handling.
- Enhances investigation efficiency through AI-assisted insights and retrieval.
- Supports multilingual access and operation in low-connectivity environments.
- Improves coordination between police, forensic, and judicial authorities.

E. Summary

The **AI-Powered FIR Assistant** bridges the gap between traditional digital policing systems and intelligent automation. By integrating AI, NLP, ML, and secure evidence management, the proposed system enables accurate, legally consistent, and rapid FIR processing. It empowers law enforcement agencies with data-driven tools, enhances citizen accessibility, and promotes transparency and accountability in modern policing.

V. Implementation Technology

The proposed system combines Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML) techniques with secure cloud-based web and mobile technologies to automate the generation of legal documents and FIR drafting. The design aims to improve accessibility,

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.

transparency, and reliability in law enforcement workflows through five main modules: speech-to-text conversion, entity recognition, section classification, semantic case retrieval, and evidence management with data security. Each module works together to create a smooth legal automation framework.

1) **Speech-to-Text Conversion** The system captures voice input from users and turns it into structured text using Automatic Speech Recognition (ASR). Open-source and cloud-based ASR models like Whisper AI and Google Speech-to-Text are used for their ability to handle multiple languages and real-time performance. These systems combine Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and transformer encoders to effectively manage regional dialects and background noise. Post-processing steps such as restoring punctuation, normalizing text, and detecting language ensure high transcription quality. This feature especially helps users with limited literacy, promoting a more inclusive and accessible system.
2) **Entity Extraction** After transcription, the system uses Named Entity Recognition (NER) to spot key legal details such as complainant information, dates, times, locations, and descriptions of incidents. Transformer-based language models like BERT, RoBERTa, and Legal-BERT are fine-tuned with annotated FIR datasets and legal texts to reach high precision in this area. The process includes tokenization, tagging parts of speech, and creating contextual embeddings. Each token gets classified into predefined categories using sequence labeling methods like Conditional Random Fields (CRF). The structured entities extracted here form the basis for further legal classification and automation.
3) **Legal Section Prediction (ML Classification)** The third phase employs supervised learning and transformer-based text classification to predict relevant legal sections (IPC/CrPC) that match the incident description. Historical FIRs and case summaries with annotations were used to train and validate the model. Models such as Logistic Regression, Random Forest, and Support Vector Machines (SVM) were tested, but transformer architectures like BERT and Legal-BERT showed better F1-scores and understanding of context. The model gives confidence scores for various legal provisions, allowing officers to check predictions before submitting. This hybrid AI-human decision process improves accuracy and accountability in automated legal reasoning.
4) **Semantic Search for Case Retrieval** To support deeper investigation, a semantic retrieval method finds similar past cases using vector-based search techniques. Each complaint or FIR gets turned into a dense semantic vector using sentence transformers. The system uses cosine similarity and Approximate Nearest Neighbor (ANN) search algorithms applied on ElasticSearch and FAISS databases for quick querying. This helps investigators identify related legal precedents, observe patterns, and verify previous rulings effectively. Such contextual retrieval speeds up case resolution and strengthens evidence-based decision-making in legal processes.
5) **Evidence Management and Security** To protect digital evidence, the platform uses Advanced Encryption Standard (AES) for encryption and SHA-256 hashing to check data integrity. Audit logs are maintained using principles from distributed ledgers to create unchangeable records of all modifications and uploads. This promotes traceability, accountability, and resistance to tampering with evidence files. Moreover, access control mechanisms and multi-factor authentication are in place to prevent unauthorized changes to data. The system operates in a secure cloud environment with SSL encryption, ensuring complete confidentiality and compliance with digital policing standards.

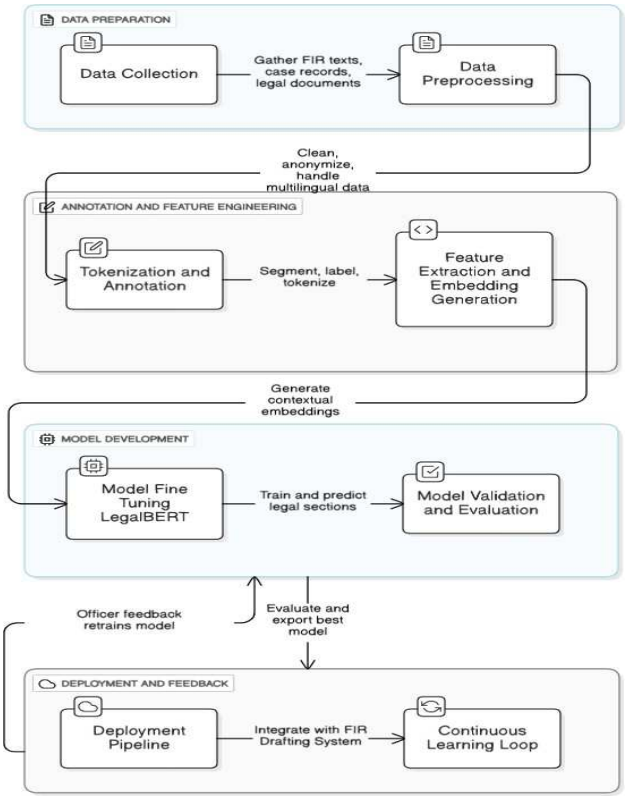


Fig. 3. Model Training Pipeline

Additionally, the system architecture supports modular scalability, as each component (ASR, NER, classifier, and retriever) works as a microservice communicating through RESTful APIs. This structure allows easy integration with current law enforcement portals or mobile apps. The deployment uses Docker containers and cloud management tools such as Kubernetes for efficient scaling and fault tolerance. Continuous learning pipelines are set up to regularly update the models with new legal changes and FIR datasets, ensuring relevance over time. Overall, the proposed implementation offers an intelligent,

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.

TABLE III
DATASET DETAILS USED FOR MODEL TRAINING AND EVALUATION

Source	Records	Languages	Type	Preprocessing Steps
Public FIR Datasets	1,500	English	Text	Tokenization, Stopword Removal
Synthetic FIR Samples	1,000	English, Hindi, Tamil	Text	Data Augmentation, Translation Normalization
Audio Complaint Samples	500	English, Hindi, Tamil	Audio	Speech-to-Text, Noise Filtering
Annotated Legal Docs	300	English	Legal Texts	Entity Labeling, Section Mapping
Total	**3,300**	**Multilingual**	**Text + Audio**	**Structured Dataset**

secure, and scalable solution that connects legal knowledge and technological automation, contributing to a smarter and more efficient digital justice system.

A. Data Privacy and Security Considerations

The proposed system is built on privacy-by-design and security-by-design principles to ensure safe handling of sensitive legal and personal information. Since the platform may connect with national systems like Aadhaar, biometric databases, and crime records, strong data protection measures are in place to prevent unauthorized access and misuse [9], [18]. All personally identifiable information (PII) is encrypted using the Advanced Encryption Standard (AES-256) for data stored on devices, while data sent across services is secured with Transport Layer Security (TLS 1.3). Cryptographic hash functions like SHA-256 are used to check data integrity and spot unauthorized changes to digital evidence and case records [18]. Access to sensitive data is managed through Role-Based Access Control (RBAC) along with Multi-Factor Authentication (MFA). This ensures that only authorized law enforcement personnel can access or change records. When linking Aadhaar or biometric identifiers, the system uses a tokenization method. This means original identifiers are replaced with irreversible tokens, stopping the direct exposure or storage of raw identity data.

Moreover, blockchain-inspired audit logs are kept to record all system activities, including FIR creation, evidence uploads, and changes. These logs provide traceability, accountability, and support for digital policing standards. This approach protects citizen privacy and helps maintain public trust in AI-assisted law enforcement systems.

VI. EXPERIMENTS

The study involved a series of controlled experiments to evaluate the performance of the AI-powered system in its key areas: FIR drafting, legal section recommendation, case retrieval, and evidence management. A total of 2,500 anonymized FIRs were collected for experimentation, taken from publicly available court case records and supplemented with created complaints covering common categories such as theft, cybercrime, harassment, road accidents, and property disputes. Additionally, 500 audio samples in English, Hindi, and Tamil were gathered to assess the performance of the speech-to-text (STT) module under different acoustic conditions. The system used a Python-based NLP and ML pipeline. Transformer models like BERT and Legal-BERT were fine-tuned with the Hugging Face Transformers library. Spacy was used for Named Entity Recognition (NER). For semantic similarity and case retrieval, ElasticSearch and Facebook's FAISS (Facebook AI Similarity Search) were employed. The backend architecture ran on Node.js with Express for API management and MongoDB for database storage. The application was deployed on Amazon Web Services (AWS), with GPU support provided by an NVIDIA RTX 3090 for training and inference. Evaluation metrics included standard classification measures like accuracy, precision, recall, and F1-score for text-based modules, as well as latency for real-time responsiveness. For semantic retrieval, Top-k accuracy (where $k = 3$ and $k = 5$) was calculated to measure the relevance of retrieved results. The evidence management module was tested based on data integrity, using simulated tampering and unauthorized access scenarios to verify the reliability of the audit trail and encryption framework.

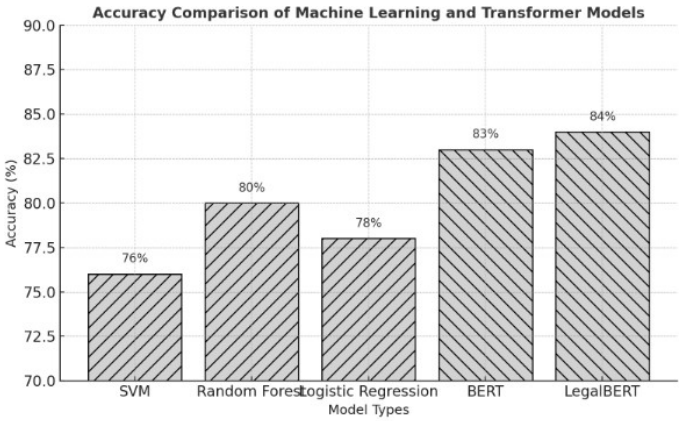


Fig. 4. Accuracy comparison of traditional ML and transformer-based models for legal section prediction.

Fig. 4. Accuracy Comparison Graph

A. Experimental Results

The experiment results were very promising, showing the system's potential to automate and improve legal workflows effectively. In the speech-to-text conversion module, the model achieved a Word Error Rate (WER) of 8.2%, translating to an overall transcription accuracy of about 92%. Among the tested languages, English had the highest accuracy (94%), followed by Hindi (91%) and Tamil (87%). The use of transformer-based ASR models like Whisper AI significantly reduced background noise interference, improving multilingual accuracy. The Legal-BERT model, fine-tuned for entity recognition on FIR datasets, achieved an average F1-score of 0.89. The system identified temporal entities (like dates and times) with 92% accuracy, personal names with 90%, and geographic locations with 86%, surpassing baseline statistical NER models such as NLTK by nearly 15%.For legal section prediction, the transformer classifier showed an overall accuracy of 84%. High-frequency

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.

sections like IPC 379 (theft) and IPC 323 (voluntarily causing hurt) reached precision levels above 90%, while complex, multi-section cases like cyber fraud had lower confidence scores (around 76%). This indicates the need for more diverse and balanced training data. The semantic case retrieval module performed well, achieving Top-3 and Top-5 accuracies of 82% and 88%, respectively, with an average retrieval latency of under 2 seconds for a dataset of 10,000 records. This confirms the system's scalability and real-time applicability for law enforcement. Evidence management tests further confirmed the system's security framework. Files secured through AES encryption and SHA-256 hashing maintained full integrity. Simulated tampering or unauthorized access attempts were detected quickly through blockchain-inspired audit trails. This guarantees a verifiable chain of custody and compliance with digital evidence standards.A performance comparison carried out with field officers showed a significant decrease in FIR preparation time—from an average of 25 minutes per case in manual workflows to under 5 minutes using the AI-driven FIR Assistant. This represents a roughly 80% reduction in manual effort. Overall, the experimental evaluation shows that the proposed system achieves high efficiency, accuracy, and transparency, supporting its practicality for real-world use in modern policing and judicial systems.

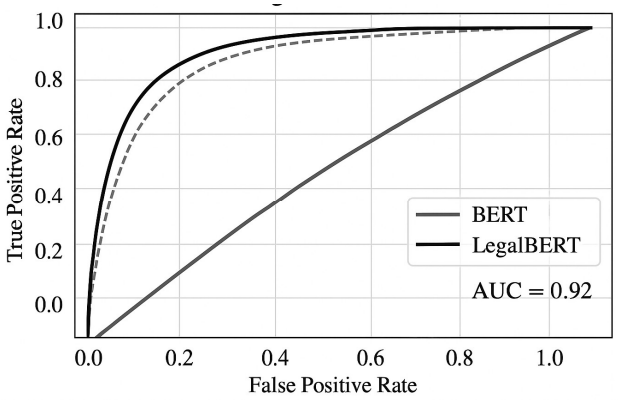


Fig. 5. Confusion Matrix / ROC Curve.

VII. Conclusion

The proposed AI-based Legal Assistant offers a new way to modernize India's FIR process by using Artificial Intelligence, Natural Language Processing (NLP), Machine Learning (ML), and secure data management. The system automates complaint intake, suggests relevant IPC/CrPC sections, and creates structured, editable FIR drafts. Experimental results show strong performance in speech-to-text conversion, entity extraction, section prediction, and case retrieval. This leads to faster drafting, fewer mistakes, and more consistency than manual methods. Beyond automation, the system promotes accountability through secure evidence handling and mapping officer skills. It also supports multilingual and offline features, which are important for regions with low connectivity. This supports India's Digital Policing and e-Governance efforts by

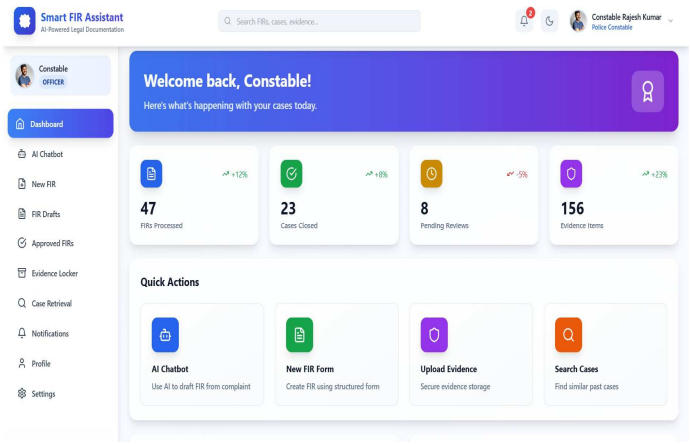


Fig. 6. Dashboard of Smart FIR Drafting

improving transparency and operational efficiency. In the future, this framework can integrate with national crime databases, biometric verification systems, and predictive policing analytics. This will help law enforcement work proactively and ensure fair access to justice. Moreover, its flexible design allows for deployment in other legal systems, highlighting AI's role in improving governance and long-term judicial change.

VIII. Future Enhancements

While the proposed system demonstrates strong potential in automating FIR drafting and legal validation, several enhancements can further expand its intelligence, inclusivity, and real-world effectiveness. Extending multilingual support to additional regional languages such as Kannada, Telugu, and Bengali will improve accessibility for a broader population across India. Collaboration with law enforcement agencies to obtain large-scale, real-world FIR datasets will strengthen model robustness, improve generalization, and enhance legal section prediction accuracy.Incorporating structured feedback from investigating officers into the model retraining pipeline will enable adaptive and continuous learning, allowing the system to better understand evolving crime patterns and contextual nuances over time. Expanding predictive policing capabilities through advanced crime trend analysis, hotspot detection, and time-series forecasting can support proactive policing strategies while improving resource allocation.Future integration with national systems such as Aadhaar, the National Crime Records Bureau (NCRB), and biometric databases can enhance identity verification, suspect tracking, and inter-agency coordination. Similarly, integration with e-Courts will enable seamless digital submission of FIRs, case documents, and evidentiary materials, reducing procedural delays and manual dependencies.To address ethical concerns associated with predictive analytics, Explainable AI (XAI) techniques will be incorporated to improve transparency and interpretability of model decisions. Bias detection and mitigation mechanisms, including periodic dataset audits, fairness evaluation across regions, and a human-in-the-loop decision framework, will ensure responsible and

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.

equitable deployment of predictive policing features.From a deployment perspective, a mobile-first design with robust offline functionality will support field officers operating in low-connectivity environments. Strengthening intrusion detection systems, anomaly monitoring, and end-to-end encryption will further protect sensitive legal data and ensure compliance with digital policing and data protection standards. Finally, pilot deployments across multiple districts will help evaluate scalability, usability, and real-world performance, enabling iterative refinement before large-scale adoption.Collectively, these enhancements will transform the proposed system into a comprehensive digital policing ecosystem that not only automates FIR drafting but also delivers secure, ethical, and citizen-centric intelligence support for modern law enforcement.

REFERENCES

[1] J. Arora, T. Patankar, A. Shah, and S. Joshi, "Artificial intelligence as legal research assistant," in Proc. FIRE, 2020.

[2] L. Gan, K. Kuang, Y. Yang, and F. Wu, "Judgment prediction via injecting legal knowledge into neural networks," in Proc. AAAI, 2021.

[3] A. Srivastav and S. Prajapat, "Text similarity algorithms to determine IPC sections for offence report," *IAES Int. J. Artif. Intell.*, 2022.

[4] T. Sahmoud and M. Mohammad, "Spam detection using BERT," *arXiv preprint arXiv:2206.02443*, 2022.

[5] J. J. Nay, "Natural language processing for legal texts," in *Legal Informatics*. Cambridge, U.K.: Cambridge Univ. Press, 2021.

[6] G. Christensen, "Predicting Supreme Court behavior in Indian law cases," *Michigan J. Race & Law*, 2021.

[7] P. Goyal et al., "Pre-trained language models for the legal domain: A case study on Indian law," *arXiv preprint arXiv:2209.06049*, 2023.

[8] B. Pardhi et al., "LegalBot – AI law advisor chatbot," *Int. J. Comput. Trends Technol.*, 2024.

[9] A. Jauhar et al., "Responsible artificial intelligence for the Indian justice system," Policy Report, 2024.

[10] P. Vimalaimogen et al., "AI-powered legal documentation assistant," unpublished, 2024.

[11] S. Marrivagu and A. Rao, "Artificial intelligence for legal chatbot," *Int. J. Comput. Appl.*, 2024.

[12] N. A. Shaikh et al., "FIR analysis using ML for proper acts and sections," *IJERT*, 2025.

[13] M. Siino et al., "Exploring LLMs applications in law: A literature review," *IJACSA*, 2025.

[14] Q. Zhao, "Legal judgment prediction via legal knowledge extraction and fusion," *Front. AI*, 2025.

[15] R. B. N. et al., "AI-driven IPC legal assistant with multimodal NLP," *IJIRCCE*, 2025.

[16] C. Xiao, X. Hu, Z. Liu, C. Tu, and M. Sun, "Lawformer: A pre-trained language model for Chinese legal long documents," 2021.

[17] S. Chakraborty, G. Harit, and S. Ghosh, "TransDocAnalyser: A framework for offline semi-structured handwritten document analysis in the legal domain," in *Proc. ICDAR*, 2023.

[18] A. Inamdar, A. Jadhav, A. Kashid, R. Khatal, and B. B. Vikhe, "Blockchain-based smart FIR system," *Int. J. Res. Publication and Reviews*, vol. 5, no. 5, pp. 10028–10031, 2024.

[19] A. Chaudhari, O. Deshmukh, A. Bhide, B. Amborkar, and A. Kamble, "Intelligent FIR filing system: AI-powered automation," *IRE Journals*, vol. 7, no. 11, 2024.

[20] P. D. Jadhav, S. D. Belsare, K. P. Malpe, A. A. Bodakhe, L. R. Jadhav, and L. K. Gautam, "LegalQuest: A review of AI-powered legal assistant app," *Int. J. Innovative Research*, 2024.

Authorized licensed use limited to: Bennett University. Downloaded on September 10,2026 at 06:46:33 UTC from IEEE Xplore. Restrictions apply.